BENJAMIN ANDERSON

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Electronic Engineer

PROJECTS

Cube
A handheld device comprising of 6 OLED screens for each face of a cube. This project required design of circuits for battery management and voltage regulation, 3D modelling of the frame, as well as efficient C code to run 6 screens and a 6 DOF sensor off a single ESP32.
`KiCAD / FreeCAD / Git / Blender`

Orthodox
A unique ortholinear keyboard. This project required rapid prototyping of custom keycaps using parametric designs, PCB coils around keyswitches to measure analog keypresses with an LDC chip, as well as a custom driver had to be written to allow for seamless integration of the keyboard with computers
`KiCAD / FreeCAD / Git / OpenSCAD`

EXPERIENCE

Support Host — **Caledonian Sleeper** 2024-2026
- Managed passenger boarding, ticket checks and payments, while responding to service issues and complaints with professionalism and tact.
- Developed strong interpersonal and conflict-resolution skills when dealing with difficult customers, resolving concerns and communicating service disruptions and alternative travel arrangements.
- Coordinated time-critical morning operations, ensuring passengers were awake, prepared and ready to disembark on schedule, while liaising with station staff to arrange breakfast orders and food deliveries.
- Responded to safety and security incidents, including medical emergencies, fire procedures and unauthorised attempts to board the train, following established procedures and contacting emergency services when required.
- Maintained clear communication with the Train Manager and passengers during significant journey disruptions, helping coordinate practical solutions in challenging situations, including remote areas of Scotland.

RF Test Engineer — **Curvalux** 2021-2022
- Used LaTeX to produce clean, formatted reports of data/graphs generated with matplotlib.
- Professionally collaborated and documented repositories using gitlab, jira, confluence and slack.
- Developed python libraries to control rotators, switches, keysight VNAs and beamforming ICs for automating radiation pattern measurements in a custom anechoic chamber.
- Translated MATLAB algorithms for phase tapering/beam steering a phased array antenna into working code on early prototype hardware.

Research & Development Engineer — **Huawei** 2019-2020
- Analysing reliability of devices with PAM4 / NRZ inspection, shear testing, electroluminescence inspection.
- Built a standalone program with PyQt to automatically produce failure analysis reports from data gathered by lab technicians, saving countless engineering hours on a repetitive manual task.
- Developed a python library to communicate with a TEC controller over the serial protocol.
- Presented weekly meetings to inform senior engineers of device failures and reliability.

Hardware Engineer — **Avalon** 2018-2019
- Tasked with designing and soldering the power management PCB for an underwater ROV
- Had meetings with suppliers to discuss partnership/sponserships for the competition.
- Regularly liaised with engineers from different disciplines to check compatibility.
- Managed my time effectively to balance commitments to University and Avalon.

REFERENCES

Jason Hanson **Princial RF Engineer**
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Xiaoli Chu **Ph.D Wireless Communications**
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Lu Zhang **Senior Reliability Engineer**
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EDUCATION

Bachelor of Engineering **University of Sheffield** 2017-2021
Electronic Engineering: 1st

A-Levels **Dunraven** 2015-2017
Mathematics B
Physics B
Computing B

GCSEs (11 Qualifications Passed) **Harris Academy** 2010-2015
Mathematics A
Science A
English B